

Discrete Structure II

CST-2202

Advanced Counting Techniques

- Solving Linear Homogeneous Recurrence Relations

Solving linear homogeneous recurrence relations

Theorem 1

- Two Distinct roots

Theorem 2

- Only one roots

Theorem 3

- k distinct roots

Theorem 4

- t distinct roots with multiplicity

Learning Outcomes

- To solve linear homogeneous recurrence relations with constant coefficients

Definition

A *linear homogeneous* recurrence relation of *degree k* with *constant coefficients* is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$.

Solving linear homogeneous recurrences

$a_n = r^n$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} \leftrightarrow r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

Solution:

Recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$

Try to find a solution of form r^n

$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$, by dividing with r^{n-k}

$r^k = c_1 r^{k-1} + c_2 r^{k-2} + \dots + c_k$, this implies

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$$

This equation is called the **characteristic equation**.

The solution of this equation are called the characteristic roots of the recurrence relation.

Theorem 1

when r_1 not equal to r_2

Let c_1 and c_2 be real numbers.

Suppose that $r^2 - c_1r - c_2 = 0$ has two distinct roots r_1 and r_2 .

Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n =$

$c_1a_{n-1} + c_2a_{n-2}$ if and only if

$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ where $n=0, 1, 2, \dots$, where α_1 and α_2 are

constants.

What is the solution of the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2} \text{ with } a_0=2 \text{ and } a_1=7?$$

Since it is linear homogeneous recurrence relation, first find its characteristic equation $r^2 - r - 2 = 0$

$$\rightarrow (r + 1)(r - 2) = 0$$

$$\rightarrow r_1 = 2, \quad r_2 = -1$$

By Theorem $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$ is a solution.

By using 2 initial conditions, $\alpha_1 = 3, \alpha_2 = -1$

$\therefore a_n = 3 \cdot 2^n - (-1)^n$ is a solution.

Theorem 2

Let c_1 and c_2 be real numbers with $c_2 \neq 0$.

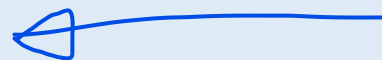
Suppose that $r^2 - c_1r - c_2 = 0$ has only one root r_0 .

A sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = \alpha_1r_0^n + \alpha_2nr_0^n$,

for $n=0, 1, 2, \dots$, where α_1 and α_2 are constants.

What's the solution of recurrence relation
 $a_n = 6a_{n-1} - 9a_{n-2}$ with initial condition $a_0=1$ and $a_1=6$?



The root of $r^2 - 6r + 9 = 0$ is $r_0 = 3$.

Hence , $a_n = \infty_1 \cdot 3^n + \infty_2 \cdot n \cdot 3^n$.

$$\because a_0 = \infty_1 = 1$$

$$a_1 = 3\infty_1 + 3\infty_2 = 6$$

$$\therefore \infty_1 = 1 \text{ and } \infty_2 = 1$$

$$a_n = 3^n + n \cdot 3^n$$

Theorem 3

Let $c_1, c_2, \dots, c_k$ be real numbers.

Suppose that $r^k - c_1 r^{k-1} - \dots - c_k = 0$

has k distinct roots $r_1, r_2, \dots, r_k$.

Then the sequence $\{a_n\}$ is a solution of the recurrence relation

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if

$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$ where $n=0, 1, 2, \dots$,

where α_1 and $\alpha_2, \dots, \alpha_k$ are constants.

Find the solution of $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$
with initial conditions $a_0=2$, $a_1=5$ and $a_2=15$.

The roots of $r^3 - 6r^2 + 11r - 6 = 0$ are

$$r_1 = 1, r_2 = 2, \text{ and } r_3 = 3$$

$$\therefore a_n = \infty_1 \cdot 1^n + \infty_2 \cdot 2^n + \infty_3 \cdot 3^n$$

$$\therefore a_0 = \infty_1 + \infty_2 + \infty_3 = 2$$

$$a_1 = \infty_1 + 2\infty_2 + 3\infty_3 = 5$$

$$a_2 = \infty_1 + 4\infty_2 + 9\infty_3 = 15$$

$$\therefore a_n = 1 - 2^n + 2 \cdot 3^n$$

e.g.

Theorem 4

Let $c_1, c_2, \dots, c_k$ be real numbers.

Suppose that $r^k - c_1 r^{k-1} - \dots - c_k = 0$ has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$ respectively,

so that $m_i \geq 1$ for $i=1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$.

Then the sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$a_n = (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n$
 $+ \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n$ where $n=0, 1, 2, \dots$, where α_1 and $\alpha_2, \dots, \alpha_k$
 are constants.

Find the solution to the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial conditions $a_0 = 1$, $a_1 = -2$ and $a_2 = -1$.

$r^3 + 3r^2 + 3r + 1 = 0$ has a single root $r_0 = -1$ of multiplicity three.

$$\therefore a_n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2) r_0^n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2)(-1)^n$$

$$\therefore a_0 = \alpha_1 = 1$$

$$a_1 = (\alpha_1 + \alpha_2 + \alpha_3) \cdot (-1) = -2$$

$$a_2 = \alpha_1 + 2\alpha_2 + 4\alpha_3 = -1$$

$$\therefore a_1 = 1, a_2 = 3, a_3 = -2$$

$$a_n = (1 + 3n - 2n^2) \cdot (-1)^n$$

Summary

To solve a linear homogeneous relation with constant coefficients,

- (1) Write the characteristic equation.
- (2) Find the characteristic value.
- (3) Write the general solution.
- (4) Find the solution using the initial conditions.

Quiz

Consider the solution of the recurrence relation $a_k = -8a_{k-1} - 15a_{k-2}$, $a_0 = 0$ & $a_1 = 2$. Which of the following is the correct relation?

A. $a_k = (-3)^k - (-5)^k$

B. $a_k = k(-3)^k - k(-5)^k$

C. $a_k = k(-3)^k - (-5)^k$

D. $a_k = (-5)^k - (-3)^k$

✓ Answer (A)

